```
conv_base.summary()

                                                                     block8_10_conv[0][0]
__________________________________________________________________________________________
conv_7b (Conv2D)                (None, 3, 3, 1536)   3194880        block8_10[0][0]
__________________________________________________________________________________________
conv_7b_bn (BatchNormalization) (None, 3, 3, 1536)   4608           conv_7b[0][0]
__________________________________________________________________________________________
conv_7b_ac (Activation)         (None, 3, 3, 1536)   0              conv_7b_bn[0][0]
==========================================================================================

Total params: 54,336,736
Trainable params: 54,276,192
Non-trainable params: 60,544
__________________________________________________________________________________________
```

Figure : Summary of InceptionResNetV2 Model

Now, we create the fully connected layers that we add on top of the downloaded model. We are going to train only this classifier, which means that after connecting the InceptionResNetV2; we will train only our classifier and freeze the InceptionResNetV2 model by the following steps:

1. First, we import layers and models from keras.
2. Then we create a Sequential model similar to our previous model.
3. Now we add conv_base (InceptionResNetV2) to the model.
4. Now we need to flatten the output from the conv_base since we want to pass it to our fully connected layer. Next, we add a dense layer to it.

```
from keras import layers
from keras import models

model = models.Sequential()
model.add(conv_base)
model.add(layers.Flatten())
model.add(layers.Dense(256, activation='relu'))
model.add(layers.Dense(1, activation='sigmoid'))  #Sigmoid function at the end bc
```

Figure : Steps to train only your classifier

5. We use the sigmoid activation function for our final layer. The last layer has just a single output that shows the probability of the image being a dog – close to one or that of a cat with a probability close to zero.

On using the model. Summary function, we find that our parameters have increased from roughly 54 million to almost 58 million. This shows